

$$p_h^2 = m_h^2 = (q + p_d)^2 = -Q^2 + 2q \cdot p_d + m_d^2$$

$$\Rightarrow Q^2 = 2q \cdot p_d + m_d^2 - m_h^2$$

$$m_h \geq 2m_p > m_d$$

$$0 > m_d^2 - m_h^2$$

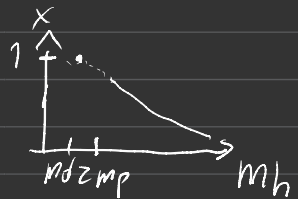
$$Q^2 < 2q \cdot p_d$$

$$x = \frac{Q^2}{2q \cdot p_d} < 1$$

Björken - x

$$S = (p_v + p_d)^2 = 2E_v m_d + m_d^2$$

$$= m_d (m_d + 2E_v)$$



$$s = (p_h + p_n)^2 = m_n^2 + m_h^2 + 2 \underbrace{p_h \cdot p_n}_{m_n \sqrt{m_h^2 + |\vec{p}|^2}} \rightarrow$$

$$s \geq (m_n + m_h)^2$$

$$\sqrt{s} - m_n \geq m_h$$

$$p_n = \begin{pmatrix} m_n \\ \vec{0} \end{pmatrix}$$

$$p_h = \begin{pmatrix} \sqrt{m_h^2 + \vec{p}^2} \\ \vec{p} \end{pmatrix}$$

$$\underline{2m_p < m_h < \sqrt{m_d(m_d + 2E_v)} - m_n}$$